Tyler Willis

Sacramento, California | Tyler.T.Willis@outlook.com | linkedin.com/in/ttwillis1/

Software Developer with 8 years of experience building modern, data-driven web applications using React, TypeScript, Python, and cloud technologies. Experienced developing public-facing applications, implementing automated testing with Playwright, collaborating across cross-functional Agile teams, and delivering maintainable, accessible software. Passionate about building intuitive user experiences and improving access to complex information through thoughtful application design.

Technical Skills

- **Programming:** TypeScript • JavaScript • Python • C# • SQL • Java • C++
- **Front-End:** React • HTML • CSS • Vite • NPM • Responsive Design
- **Testing:** Playwright • Automated Regression Testing • End-to-End Testing
- **Cloud / DevOps:** AWS • Docker • Azure • Git
- **Additional:** REST APIs • Agile • Scrum • Accessibility (WCAG) • Drupal

Professional Experience

Software Engineer III

Public Consulting Group, March 2021 – Present

- Developed and maintained modern **React and TypeScript** applications integrating REST APIs and cloud services across AWS and Azure.
- Built comprehensive automated end-to-end testing suites using **Playwright**, improving release quality by identifying dozens of defects before production.
- Led Independent Verification & Validation (IV&V) efforts, improving software quality, reliability, and compliance across multiple enterprise projects.
- Supported application modernization efforts using Microsoft Power Platform and cloud technologies.

Software Engineer

Pacific Northwest National Laboratory, September 2019 – February 2021

- Developed and maintained **public-facing Drupal websites** used across multiple scientific research organizations.
- Built reusable themes and interface components while collaborating with designers and stakeholders.
- Performed security updates and maintenance across more than **70 production websites**, ensuring reliability and long-term sustainability.

Tech Intern IV

Pacific Northwest National Laboratory, May 2019 – September 2019

- Supported the Drupal team by streamlining development processes and executing backlog tasks.
- Achieved full-time developer status due to outstanding performance.

Android Application Developer

Benton County Mosquito Control, January 2019 – May 2019

- Designed and developed “Mosquito MD Abatement”—an offline Android app with custom map tile rendering.
- Improved pesticide tracking and reporting efficiency for field technicians.

Key Projects

CA DHCS PATH

- Designed and implemented a comprehensive Playwright regression testing framework that identified dozens of defects across multiple production deployments.

CA Department of Fi$Cal

- Led an IV&V readiness assessment project as Project Owner and Scrum Master.

CA Vaccine Rollout

- Contributed to development of California’s COVID-19 vaccine distribution system.

NE Sensor (U.S. Department of Energy)

- Designed and deployed a web platform for managing scientific sensor data, emphasizing usability, data visualization, and reliable information access.

Freelance 3D Artwork & Automation

- Created custom 3D assets using Blender and automated bulk GIF generation with Python.

Mosquito MD Abatement

- Developed an Android app that integrates offline functionality and custom map tile rendering for field operations.

Education

Bachelor of Science - Software Engineering
Brigham Young University - Idaho
September 2014 – July 2019

Focus Areas: Graphics, Vector Mathematics, Video Game Development